E30 ("invariants", lite): an accounting audit of trivial $\mathbb{F}_{p^4}$ products

Run report and results, 2026-10-30. Note §9.28 (pre-registration, `b3ddccb13`) and §9.32 (results). Code and binary `0e4fccce1`. Records `experiments/35_invariants_lite/`. Host: the 4-vCPU container; no isolation tier; operation counts only. **No Stage B.**

What E30 is. The lost draft's counter change could not be reconstructed (the note records neither the counter nor the draft). Registered and run instead: a separate counter that re-prices products whose operand is 0 or 1 (to 0) or lies in $\mathbb{F}_p$ (to 4, a scaling). Recorded counts do not change.

Measured

- IC side (E27′s Karatsuba solver-A runs plus the audit, 20 runs, 0 failures): every non-count field equal to E27′s records; trivial products 0.3087–0.3131 % of S.
- Walks (E25′s seeds, 320 walks): saving 0–1.1 $\mathbb{F}_p$ multiplications per walk (≤ 0.0002 %); the GLS multipliers are not in $\mathbb{F}_p$.
- Distance to the matched $\langle \pm\psi \rangle$ walk (E27 (iii)): −0.309 to −0.313 % at 0 GAE, −0.084 to −0.086 % at 0.1, −0.011 % at 1 GAE.

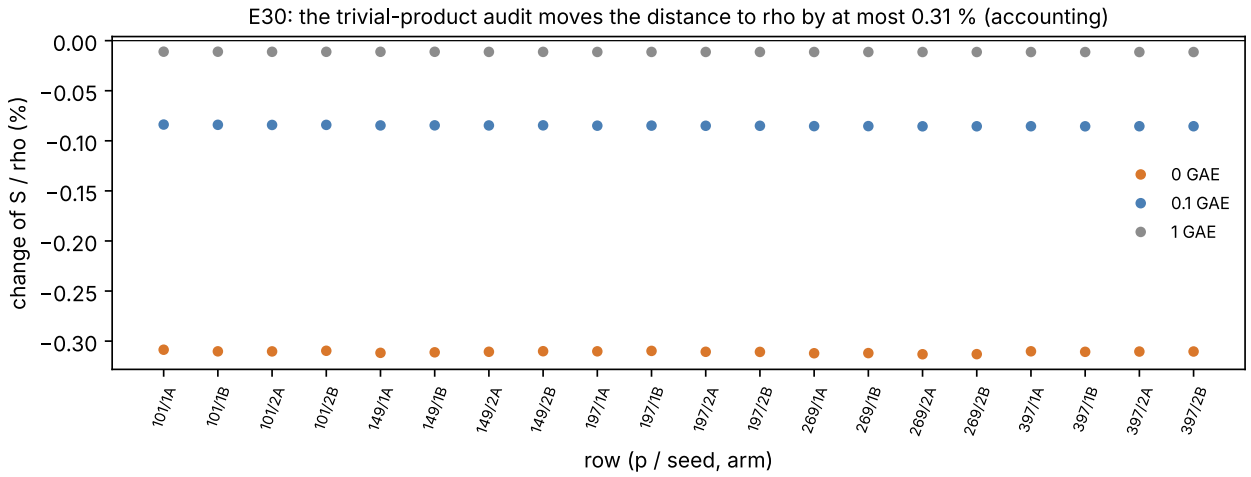

Figure 1: Change of S/rho per row under the audit at three weights. Source: `summary.json`.

Verdicts (§9.28)

G2.110, G2.111, G2.112, G2.113 hold (largest change 0.313 %; the red team's estimate for the lost draft was ≈ 0.6 %).

What it says

Accounting, not an improvement: no algorithm changed. Recorded so that a counter change of this size is not read as progress. Main's 2026-10-24 change (charging the walks' $[d]G$ check, §9.29) moved the same distances by 0.13–0.78 % the other way.